Evaluation of scenario reduction methods for stochastic inflow in hydro scheduling models

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***Abstract*—The long-term hydropower scheduling problem is inherently stochastic due to uncertainty in future reservoir inflow. We use Stochastic Dual Dynamic Programming (SDDP) to solve this problem. This work evaluate and compare three scenario reduction methods used to construct a multistage scenario tree which represents the underlying stochastic inflow process in the SDDP model. A case study is carried out to numerically assess the performance of the different scenario reduction methods. The performance is measured using out-of-sample simulation, simulating the solution strategies obtain with the various scenario models on an exogenously given set of inflow scenarios. Our results show that the choice of scenario reduction method impacts the solution to a hydropower operation planning problem substantially.**

***Index Terms*—Hydropower, scenario reduction, stochastic dual dynamic programming, stochastic inflow**

I. Introduction

The long-term (1-5 years) hydropower scheduling problem is important for systems containing large reservoirs with multiannual regulation capabilities. Results from long-term models are used for several purposes, including risk analysis, maintenance scheduling and for making various predictions. Moreover, the models provide target values for short-term scheduling and daily bidding processes [1]. Due to substantial uncertainty related to important parameters, in particular reservoir inflow, it is usually considered more important to properly account for uncertainty than to include a high level of technical details in long-term models [2]. We consider operation planning problems which can be formulated as a multistage stochastic linear programming problem and solved using Stochastic Dual Dynamic Programming (SDDP) [3], [4].

Solution methods for stochastic problems, including SDDP, require the stochastic parameters to be represented by a finite number of scenarios. A scenario is one possible outcome of the random data, with a corresponding probability. Moreover, the number of scenarios must be relatively small in order for the stochastic model to be computationally tractable. There exists numerous approaches to scenario generation. We refer to [5] and reference therein for an overview and discussion of different scenario generation methods. Related to scenario generation is scenario reduction methods. The goal of scenario reduction is to find a scenario subset of prescribed cardinality that provides a good approximation to the initial scenario set. Methods based on minimizing a probability metric [6], [7] are popular in stochastic programming. Scenario reduction can also be accomplished by clustering methods.

Previous work on scenario generation and reduction in applications for hydropower scheduling include [8]–[10]. The problem of optimal scenario tree reduction for stochastic streamflows was addressed in [8]. They present two methods that are assessed numerically on inflow sequences generated for the Brazilian power system. [9] analyzed four different clustering techniques used to generate a scenario tree, which subsequently was reduced. The methods were applied to hydro inflow and compared. [10] apply k-means clustering to an initial scenario set generated from a statistical model, to construct a scenario tree. The sensitivity and stability, with respect to the scenario tree, of scheduling policies resulting from solving a two-stage hydro-thermal scheduling problem was analyzed. Both [9] and [8] evaluate and compare different methods, but the quality was not assessed in any applications for production planning.

SDDP is a solution algorithm that utilizes sampling to keep the problem computationally tractable. Most of the research concerning inflow modelling in SDDP-based operation planning models has focused on different sampling strategies to internally generate inflow scenarios [11], [12]. This work concentrates on different scenario reduction procedures which can be used to construct a scenario tree, representing stochastic inflow in SDDP based scheduling models.

The basis for generation of inflow scenarios in this work is a statistical time series model, which account for seasonal effects and autocorrelation. A white noise process constitute the random component of the model. Because the noise is stage-wise independent, methods for scenario generation and reduction developed for two-stage problems apply [13]. Our approach to construct a scenario tree combines a stage-wise discretization of the noise distribution with the use of a parametric description of the underlying inflow process. A scenario tree can then be generated from the inflow model by considering all the discrete noise alternatives in each stage. This scenario representation within SDDP was used in [12], where the scenario reduction was carried out using K-means clustering. In this work, we analyze and compare three different methods to obtain the noise scenarios; K-means

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clustering [14], fast forward selection [6], [7] and a simple approach called the 'bracket mean' method [15].

We restrict attention to approximation of a univariate stochastic process. In practice, the process is often a multivariate one and often of large dimension. This have commonly been handled using principal component analysis, e.g. [1], to reduce the dimension. Scenario reduction can then be carried out on each principal component individually, and the resulting scenarios are combined all against all to form a discrete approximation to the multivariate distribution. Thus, the methods and analysis provided here are also relevant for multivariate stochastic processes.

The quality of the scenario reduction method should be measured in terms of the solution to the underlying optimization problem [16]. One of the contribution of this paper is therefore to increase the understanding about how different scenario representations may influence the quality of the solution to the hydropower planning problem. To this end, a realistic case study was carried out to evaluate and compare different scenario representations for the stochastic inflow parameter. We compare the quality of the obtained solutions using different scenario models, derived from the same initial scenario set, by out-of-sample simulation. With this approach, the resulting solution strategies are simulated on a different sample than was used to solve the problem, and thus provides a neutral basis for comparison of the different methods.

The remainder of this paper is organized as follows. Section II briefly describes the structure of the long-term planning problem, the SDDP solution methodology and how inflow uncertainty is represented within this framework. The different scenario reduction methods used in this study are outlined in section III. Section IV presents numerical results, and finally section V ends the paper with a discussion and suggestions for further work.

II. Problem formulation and SDDP

The goal of long-term hydro planning for a producer in a deregulated market is to determine a weekly operation schedule which maximizes expected profits over the planning period (risk-neutral approach). A high level of technical details is usually omitted in long-term models and nonlinear relationships are usually approximated by linear or piecewise linear expressions, e.g. [2]. The planning problem can then be formulated as a multistage stochastic linear programming problem.

The long-term hydro scheduling problem can in principle be solved by stochastic dynamic programming or scenario-based approaches such as (nested) Benders decomposition. However, they both suffer from the so-called *curse of dimensionality*, see e.g. [17]. SDDP [3], [4] was developed to overcome these dimensionality problems. It is an iterative algorithm that employs a combination of sampling and Benders decomposition [18]. Instead of using a complete pre-generated scenario tree, the SDDP method utilize internal sampling, sampling the scenarios during the solution procedure. The technique of Benders decomposition is used to find a set of cuts that constructs a linear approximation to the expected future benefit function in each stage.

One main iteration of the SDDP algorithm consists of a forward simulation and a backward recursion:

- **Forward pass** The purpose of the forward pass is to generate trial solutions that yield reasonable values for the state variables used in the next backward pass. For each sampled scenario, a set of linear programming problems is solved at each stage, moving forward to the last stage. End reservoir levels and costs at each stage are saved. At the end of the forward pass, the total cost for each scenario is calculated and a convergence criterion is tested. If convergence is not achieved, the algorithm starts the backward pass. The scenarios are resampled in each forward pass.
- **Backward pass** The solutions are updated by adding cuts that approximate the expected future cost function to each stage problem, starting at the last stage and moving backward to the first. At each stage, given the values for the state variables found in the last forward pass, the next stage problem is solved for all possible outcomes of the stochastic parameters (for the current stage). The expected values of the cost and dual variables are saved and used to construct a cut that is passed to the previous stage problem.

Different sampling strategies have been considered for SDDP [11], [12]. The approach taken in this work follows the one described in [12], referred to as selective sampling. With this approach, the basis for the internally sampled scenarios is a complete finite scenario tree which is assumed to represent the true process adequately.

Figure 1 illustrates the sampling process for one iteration of the algorithm. Imagine the stochastic inflow process is fully represented by the complete scenario tree in Figure 1a. Figure 1b illustrates the forward simulation pass, where a predetermined number of scenarios is sampled from the complete tree and a sequence of one-stage problems are solved for each sampled scenario. Note that, to ensure convergence of the algorithm, the forward scenarios should be resampled in each iteration [19]. The backward pass traces back along each of the sampled forward scenarios and solves the next stage problem for each branch leaving the node, as shown in Figure 1c.

III. Scenario tree generation

This section presents the stochastic inflow model which constitute the basis for the scenario tree generation, and the scenario reduction methods used to construct a tree of manageable size.

A. Stochastic inflow model

We use a linear time series model accounting for seasonality and lag-one autocorrelation. This model is commonly used in SDDP based scheduling models in the Nordic countries [1].

Weekly inflow series are usually characterized by seasonal variations and is as such a non-stationary process. The first

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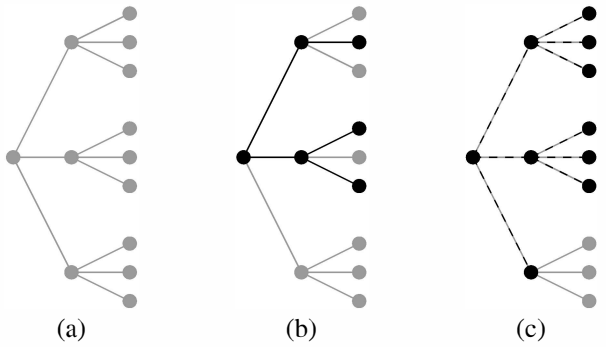


Fig. 1. Illustration of scenario representation of stochastic inflow in the SDDP algorithm for a three stage problem. The first stage is deterministic. (a) the complete scenario tree, (b) scenarios sampled in the forward pass and (c) scenarios solved in the backward pass

step is therefore to eliminate seasonal variations such that we are left with a seasonally adjusted time series which can be considered (weakly) stationary, and further described by a non-seasonal AR(1)-process. Let a_t denote a univariate seasonal inflow time series. The original series a_t is deseasonalized by subtracting the weekly mean and standard deviation,

$$z_t = \frac{a_t - \mu_s}{\sigma_s}, \tag{1}$$

where the subscript notation s represents the season corresponding to the time index t. For weekly data, the relationship between the season and time index is given by $s = (t-1) \mod (52) + 1$.

An estimate of the mean of season s across all of the years M is determined using

$$\hat{\mu}_s = \frac{1}{M}\sum_{i=1}^{M} a_{s,i}, \qquad s = 1, 2, ..., 52,$$

and, correspondingly, the standard deviation of season s is estimated by

$$\hat{\sigma}_s = \sqrt{\frac{1}{M}\sum_{i=1}^{M}(a_{s,i} - \hat{\mu}_s)^2}, \qquad s = 1, 2, ..., 52.$$

Subsequent to deseasonalization, the series z_t is fitted to an AR(1) model

$$z_t = \phi z_{t-1} + \epsilon_t, \tag{2}$$

where ϕ is the autoregressive parameter. The innovation process ϵ_t is assumed to be a white noise process, that is, it is uncorrelated with zero mean and constant variance.

B. Scenario reduction

The aim of scenario reduction techniques is to downsize an original scenario set, while keeping the loss of stochastic information embedded in it as small as possible. Since the random error terms are stage-wise independent, scenario reduction methods for two-stage problems can be applied to each stage separately. The problem of scenario reduction thus reduces to that of discretizing the the original error distribution into K discrete outcomes $\{\epsilon^k\}_{k=1}^K$ with corresponding probabilities $p_k \geq 0$, $k = 1, ..., K$. A full scenario tree can then be built by evolving the stochastic inflow model, using the full discrete noise distribution in each stage.

We consider three different scenario reduction methods which are outlined in the following.

1) Fast forward selection: Consider a probability distribution P defined over an initial (large) scenario set Ω with cardinality N. In the case where Ω is obtained by simple random sampling from a specified distribution, the scenarios have equal probabilities $p_i = 1/N, i = 1, 2, ..., N$. The objective is to find the subset $\Omega_{\text{red}} \subset \Omega$ of cardinality K and assign to it new probabilities, such that the corresponding approximate distribution Q is closest to P with respect to a probability distance. We refer to [6], [7] for details on the notion of optimal reduction with respect to probability metrics.

The most commonly used probability distance in stochastic programming is the Kantorovich distance. Let $\omega \in \Omega$ and $\tilde{\omega} \in \Omega_{\text{red}}$ denote scenarios belonging to their respective scenario sets. For discrete distributions the Kantorovich distance D_K between the probability distributions P and Q can be determined by

$$d_K(P, Q) = \min \sum_{\omega \in \Omega \setminus \Omega_{\text{red}}} p_\omega \min_{\tilde{\omega} \in \Omega_{\text{red}}} d(\omega, \tilde{\omega}), \tag{3}$$

where p_ω is the probability of scenario ω, and $d(\cdot)$ denotes the Euclidean distance.

Based on this expression, several heuristic algorithms have been developed to find Ω_{red}, including the FFS algorithm introduced in [6] and further refined in [7]. FFS is appropriate when the number of scenarios in the reduced set is much smaller than the number of initial scenarios (strong reduction). It is a greedy algorithm that starts with an empty set, and iteratively selects the scenario that minimizes the Kantorovich distance between the reduced and initial scenario set, and adds it to the reduced scenario set. The algorithm ends when a predetermined number of scenarios have been selected and new probabilities are assigned to them. The new probability of a selected scenario is equal to its former probability plus the probabilities of deleted scenarios that are closest to it with respect to $d(\cdot)$.

2) K-means clustering: K-means clustering [14] is a well-known heuristic clustering algorithm which has been applied for scenario reduction purposes [10], [10], [12], [20]. The method partitions the initial scenario set into K disjoint clusters. Each cluster is described by its center and members, found by minimizing the sum of distances from each cluster member to its center, over all clusters. The distance measure used in this work is the squared Euclidean distance. This algorithm moves objects, i.e. the initial scenario values, between clusters until the sum of distances cannot be decreased any further. The cluster centers form the reduced scenario set and inherits the total probability of its cluster members. The

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probability of scenario k is given by

$$p_k = \frac{N_k}{N},$$

where N is the cardinality of the initial scenario set and N_k is the number of objects allocated to cluster k.

3) Bracket mean: The 'bracket mean' (B-mean) method is a simple approach to reduce an initial scenario set of cardinality N into a reduced set of smaller cardinality K. The initial scenario set is first sorted and then divided into K intervals, where each interval contains N/K scenarios. In the case where N is not divisible by K, the additional scenarios are included in the first interval. The mean of each interval is calculated and form the reduced scenario set. All the scenarios in the reduced set have probability $1/K$.

IV. Results

In this section, the three different scenario reduction methods outlined in section III-B are evaluated and compared. The quality of the reduction methods are examined from two perspectives. First, the statistical properties of the approximated discrete distribution are compared to those of the original distribution. Secondly, the quality of the different scenario models are evaluated through a realistic case study.

A modified version of the Tokke hydropower system located in the south of Norway was used in the case study. The system consist of seven interconnected reservoirs and power plants, and has a total installed capacity of 986 MW. Inflow information for this site is contained in the univariate series 'Eggedal', for which weekly observations from 1958 to 2013 are available from the Norwegian Water and Resource Directorate (www.nve.no).

A. Statistical results

The parameters of the inflow model was estimated using 2912 weekly observations for the inflow series 'Eggedal'. A usual diagnostics checking was carried out and showed that the errors are uncorrelated with approximately constant mean and variance. However, the noise distribution is right-skewed and clearly deviates from normality. For the purpose of this study we assume that the empirical distribution $\{\hat{\epsilon}_t\}_{t=1}^{T}$ provides a reasonable approximation to the true error distribution and let it constitute the original scenario set.

The original scenario set was subsequently reduced into K scenarios using each of the reduction methods presented in section III-B. We considered $K = 3$ and $K = 7$ scenarios. Figure 2 and 3 graphically display the resulting discrete distributions for $K = 3$ and $K = 7$ scenarios (blue bars), respectively, together with the normalized histogram of the initial scenario set (light grey). It is clear from these plots that the scenarios and their probabilities obtained differ between the reduction methods. In particular, the scenarios obtained using K-means covers a wider range of the original noise distribution.

Table I reports the mean, standard deviation and skewness of the original and the approximated distributions discretized into $K = 3$ and $K = 7$ scenarios.

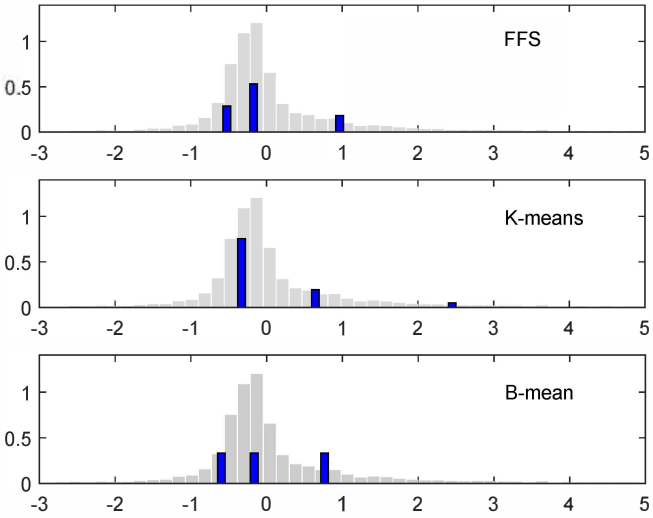


Fig. 2. Approximate distribution with $K = 3$ scenarios (blue bars). The normalized histogram for the initial distribution is shown in light grey.

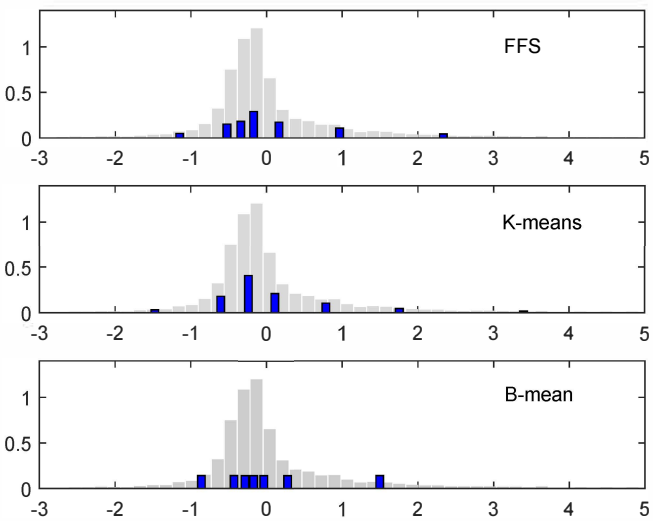


Fig. 3. Approximate distribution with $K = 7$ scenarios (blue bars). The normalized histogram for the initial distribution is shown in light grey.

TABLE I
MEAN, STANDARD DEVIATION AND SKEWNESS FOR THE ORIGINAL AND APPROXIMATED ERROR DISTRIBUTIONS.

Statistic	Original	FFS	K-means	B-mean
		$K = 3$		
Mean	0.000	-0.006	0.000	0.000
Std.	0.789	0.510	0.684	0.568
Skewness	2.029	2.507	15.592	0.412
		$K = 7$		
Mean	0.000	-0.001	0.000	0.000
Std.	0.789	0.694	0.764	0.692
Skewness	2.029	5.259	13.554	1.134

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We see that the mean is preserved by the approximate distributions obtained by the k-means and b-mean reduction methods, while it is slightly lower than that of the original scenario set for the FFS method. However, it approaches the nominal value when the number of scenarios is increased to seven. In the case of three scenarios, all methods yield a standard deviation that is lower than for the original distribution. The standard deviation increases when the number of scenarios is increased from 3 to 7, for all methods. However, the K-means method yields a standard deviation that is higher, and closer to that of the original distribution, than the FFS and B-mean methods. Since the original noise distribution is not symmetric, we also compared the skewness (Fisher-Pearson coefficient of skewness). While the deviation from the original distribution is moderate for the discrete distributions obtained by FFS and B-means, the skewness is substantially higher in the scenarios obtained by the K-means method.

B. Application to a hydropower scheduling problem

A simple scheduling problem for the test system was considered to numerically assess the performance of different scenario reduction methods.

Decisions are taken weekly over a planning horizon of one year, and the planning period starts in the first week of January. The producer operates in a deregulated electricity market, and is small enough to be considered a price-taker. The objective of the modelling is to find a release policy that maximizes expected revenues from sales in the day-ahead market, minus penalty costs for violating bounds on the reservoir volumes, over the planning horizon. For the purpose of this analysis we assume the market is represented by an exogenously given deterministic price. Since the price is deterministic, the SDDP method applies directly to this problem, whereas a stochastic price would require a hybrid SDDP based solution approach as described in [1]. The constraints of the problem are water balance equations and bounds on reservoir volume and water discharge, for individual reservoirs and plants.

The inflow scenario models were preprocessed in MATLAB, and integrated with an SDDP model developed in C++ for the hydro scheduling problem. To limit computation time we stopped the algorithm after 15 iterations, which was sufficient for the upper bound to stabilize. We used 100 sampled scenarios in each forward pass and these were resampled in each iteration. The number of backward openings was equal to the number of branches in the complete scenario tree; $K = 3$ and $K = 7$.

We compared the different reduction methods using an out-of-sample simulator approach. This was carried out as follows: After the last iteration of the SDDP algorithm a final forward simulation was performed to evaluate the resulting policy, using a new set of realistic inflow scenarios. It is important that these scenarios are provided exogenously, and to this end we used 1000 scenarios generated from the estimated inflow model, by sampling from the *original scenario set* of the noise distribution.

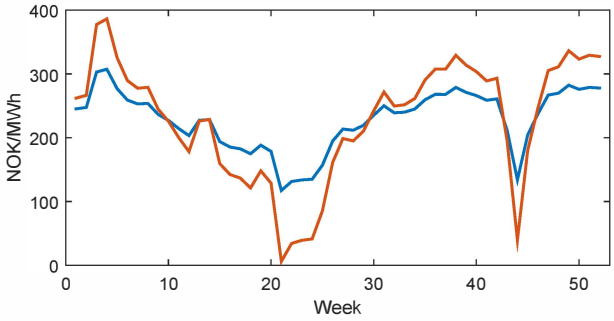


Fig. 4. Average weekly day-ahead prices from 2014 (blue) and average weekly prices with inflated variance (red)

We considered two different price series. The first series are weekly average Nord Pool day-ahead prices for price area NO1 from the year 2014, published by the Nordic power exchange Nord Pool (www.nordpoolspot.com). The second price series was constructed by artificially inflating the variance of the first series, while retaining the yearly average. Both price series are displayed in Figure (4).

For each of the two price series, the scheduling model was run for six cases, differing only in the scenario representation of stochastic inflow. The performance is measured in terms of the expected value of the objective function from the out-of-sample simulation. Higher values are better. The scenarios used in the simulation was pregenerated and equal for all cases and price profiles.

TABLE II
EXPECTED VALUE OF OBJECTIVE FUNCTION (MNOK) FOR OUT-OF-SAMPLE SIMULATION WITH 1000 SYNTHETIC INFLOW SCENARIOS. ORIGINAL PRICE SERIES.

K	FFS	K-means	B-mean
3	1298.79	1289.29	1298.47
7	1297.85	1297.47	1298.68

Table II shows the simulation results obtained with the original price series. The expected values of the objective function are very similar for all the scenario models, except for the K-means clustering with $K = 3$ which performs noticeable worse than the other scenario models. The performance of K-means improves with more than 8 MNOK when $K = 7$, for which it is approximately equal to the other models.

The simulation results from using the variance inflated price series are shown in Table III. We observe that the results are consistent with the ones obtained using the original price. The

TABLE III
EXPECTED VALUE OF OBJECTIVE FUNCTION (MNOK) FOR OUT-OF-SAMPLE SIMULATION WITH 1000 SYNTHETIC INFLOW SCENARIOS. VARIANCE INFLATED PRICE SERIES.

K	FFS	K-means	B-mean
3	1392.39	1378.42	1392.29
7	1392.03	1388.68	1390.78

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worst scenario model is still the one obtained with K-means clustering for $K = 3$. Although the result for this method improves by more than 10 MNOK for the case with $K = 7$, it still performs worse than the FFS and B-mean methods. Thus, the difference in performance between K-means and the other methods increase when we use a more variable price. This was expected, since the potential benefits of a more efficient use of the available water will increase with increasing price variations.

Overall, the results from this study have shown that it is difficult to select a unanimous 'winner' among the scenario reduction methods. FFS and B-mean performs approximately equally well for both $K = 3$ and $K = 7$, given either of the price series. However, the study did reveal that the K-means method is not appropriate compared to the other scenario reduction methods for this hydro scheduling problem.

Finally, a comment is in order with regards to the results for the B-mean reduction method, using the variance inflated price. One would not expect that the case with three scenarios performs better than the case with seven scenarios. We repeated the model run and out-of-sample simulation, using a new set of 1000 inflow scenarios in the simulation part. The results showed that the case with $K = 7$ now achieved an expected value of the objective function that was 1.64 MNOK higher than the case with $K = 3$. It is therefore likely that differences of this magnitude can be attributed to sampling errors.

V. Discussion

In this paper, we have evaluated and compared three different methods for scenario reduction used to provide a scenario tree, which can be sampled from during the solution procedure of SDDP.

A case study was carried out to numerically assess the performance of the different scenario reduction methods. Out-of-sample simulation was used to compare the solution strategies obtained using the different scenario models. This study demonstrated that the choice of scenario reduction method affect the solution of the multistage hydropower scheduling problem solved using SDDP. Of the three methods investigated in this work, the FFS and B-mean methods performed substantially better than the K-means clustering approach. Although we cannot conclude that one method is better than another for a general hydropower planning problem based on this study, it is interesting to observe that different methods affect the solution strategy and in turn the simulated expected net income. In particular, the K-means clustering method performs substantially worse, and was much more sensitive to the number of scenarios, than the other two scenario reduction methods considered here.

Further work is under way to investigate the sensitivity of the results with respect the number of forward scenarios sampled and to the out-of-sample scenario set, for which the obtained solution strategies are simulated on. A more thorough performance analysis should be carried out by considering different test systems and operation planning designs, in order to draw more solid conclusions about the quality of the different scenario reduction methods.

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